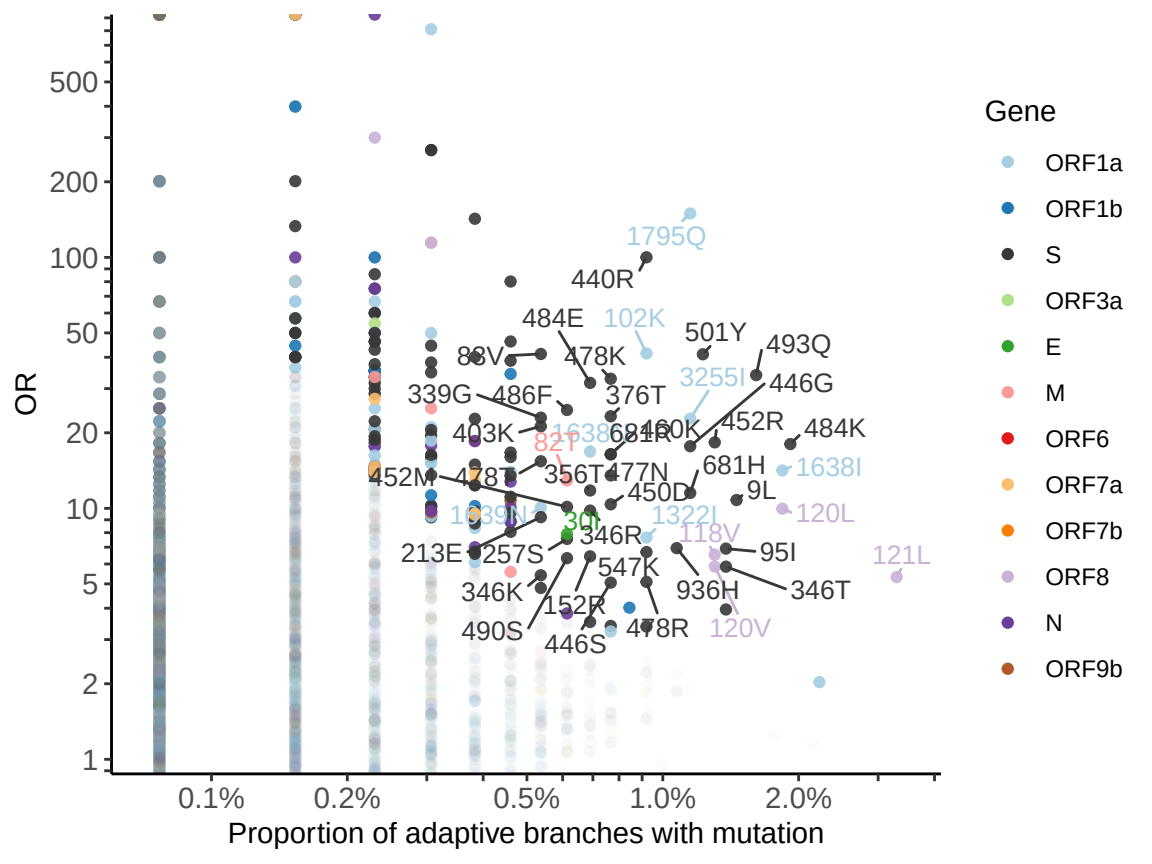

OR
500
200
100
50
20
10
5
2
1
0.1%
0.2%
0.5%
1.0%
2.0%
Proportion of adaptive branches with mutation
1795Q
440R
484E
102K
501Y
493Q
86V
478K
3255I
446G
339G
486F
376T
452R
484K
403K
1638I
681R
460K
821T
681H
1638I
452M
478T
356T
477N
450D
9L
1639N
301I
1322I
120L
346R
118V
213E
257S
547K
95I
121L
346K
936H
346T
152R
490S
478R
120V
446S
Gene
ORF1a
ORF1b
S
ORF3a
E
M
ORF6
ORF7a
ORF7b
ORF8
N
ORF9b